

PROJECT REPORT

SB WORKS

(FREELANCE FINDER PLATFORM)

SUBJECT: Full Stack Software Development

MODULE: MERN Stack Implementation

Under the Guidance of
Department of Engineering & Research

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1. INTRODUCTION

1.1 Project Overview

The project SB Works, conceptually referred to as FreelanceFinder, is a multi-dimensional digital marketplace platform. In the wake of the global shift toward remote work and the decentralized gig economy, there arose a critical need for a platform that simplifies the nuances of client-freelancer collaboration.

SB Works is designed to be a revolutionary platform that moves beyond the typical bulletin-board style listing services. Instead, it offers a managed ecosystem where professional integrity, real-time communication, and secure work submission are the cornerstones of every interaction.

Built using the MERN stack—comprising MongoDB, Express.js, React, and Node.js—the platform ensures that users experience zero latency. Whether a client is looking for a graphic designer for a local bakery or a team of developers for a technical overhaul, SB Works provides the administrative oversight and intuitive interface required to make the project a success.

The platform is not merely a tool for established professionals; it is particularly engineered to help recent graduates and entry-level specialists build their careers through a transparent bidding and rating mechanism that values current work quality over long-term history alone.

1.2 Purpose

The primary purpose of SB Works is to bridge the chasm between untapped talent and local business needs. We identified that many local businesses, such as "Sugar Rush" (our bakery case study), struggle to find accessible creative talent because traditional platforms are often too globalized or complex for small projects.

The second purpose is to provide a "Safe Harbor" for communication. When freelancers and clients take their conversations off-platform to email or private chat, data security and payment integrity are compromised. SB Works internalizes all collaboration to ensure that both Sarah and the Bakery have a documented history of project milestones and design directions.

Thirdly, the purpose is educational and career-driven. By giving users a dashboard that reflects their progress and reviews, we empower individuals to treat freelancing as a professional business entity.

2. IDEATION PHASE

2.1 Problem Statement

The fundamental problem addressed by SB Works is the Transparency and Reliability Gap in the freelance market. Our research into the experiences of university graduates showed that while they possess the skills (HTML, CSS, React, etc.), they lack the platform to prove them without being drowned out by larger agencies.

Additionally, clients often report a "Communication Breakdown." When requirements are updated during a logo redesign project, for instance, those updates often get lost. There is no central source of truth for the project state. SB Works solves this by creating a dedicated, bid-associated workspace for every single project posted on the platform.

2.2 Empathy Map Canvas

The empathy mapping process focused on our core personas: The Eager Freelancer (Sarah) and The Busy Client (Local Business Owner).

[FIGURE 2.1: EMPATHY MAP CANVAS - THE FREELANCER]

SENSES: Sees high-competition on global sites. Hears peers struggling to find work.

THINKS: "I am a great designer, but I have no reviews yet. How do I start?"

FEELS: Motivated but anxious about the first client interaction.

PAINS: Rejection of bids, high platform fees, complex interfaces.

GAINS: A glowing review, a professional portfolio, and a repeat client.

2.3 Brainstorming

During the brainstorming phase, we utilized several techniques including "Mind Mapping" and "Worst Possible Idea" to identify potential threats to the platform's integrity.

Brainstormed Feature	Priority	Reasoning
Integrated Chat System	High	Prevents data loss and off-platform fraud.
Portfolio-in-a-Bid	High	Allows Sarah to attach samples directly to the proposal.
Admin Verification	Medium	Ensures "Sugar Rush" bakery is a legitimate entity.
One-Click Review	High	Simplifies the feedback process for busy clients.

Real-time Notification	Critical	Informs freelancers instantly when their bid is selected.
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This phase concluded that simplicity in the UI (User Interface) would be the biggest differentiator from existing overcrowded platforms.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

A detailed analysis of the user journey was performed to ensure that from registration to project completion, the user feels "Discovering Opportunities" and "Unlocking Potential."

[FIGURE 3.1: CUSTOMER JOURNEY MAP]

1. Awareness: Sarah finds SB Works via a social media search for creative gigs.
2. Research: Sarah browses "Logo Design" categories and finds the Bakery project.
3. Decision: Sarah submits a bid with her React-based designer portfolio.
4. Working: Sarah chats with the bakery owner to refine the logo color palette.
5. Advocacy: Sarah receives a 5-star review and shares her success story.

3.2 Solution Requirement

The requirements are split into Hardware, Software, and System functionality.

3.2.1 Functional Requirements (The "What")

- The system **MUST** allow users to register as either a Client or a Freelancer.
- The system **MUST** provide a secure file upload for project deliverables.
- Clients **MUST** be able to filter bidders based on their relevant skill tags.
- The Admin module **MUST** be able to suspend malicious accounts.

3.2.2 Quality Requirements (The "How Well")

- Security: All communication sessions must be JWT-authenticated.
- Performance: API endpoints must process requests in under 200ms.
- User Experience: Minimum 90/100 Lighthouse score for accessibility.

3.3 Data Flow Diagram (DFD)

The Data Flow Diagram represents how information is processed through the SB Works backend architecture.

[FIGURE 3.2: DFD LEVEL 1 - DATA PERSISTENCE]

Input: Project Details Form -> Process: Express Controller -> Logic: Validate

Schema -> Storage: MongoDB Cluster

External Entity (Sarah) -> Query: Category Filter -> Output: Real-time Project
List

The flow ensures that any data entered by the client (Post project) is sanitized before being written to MongoDB, preventing NoSQL injection attacks.

3.4 Technology Stack

The MERN stack was chosen to handle the real-time demands of SB Works.

1. MongoDB: A NoSQL database storing user profiles and project documents.
2. Express.js: The backend framework providing REST API connectivity.
3. React.js: The frontend library managing component-based UI and state.
4. Node.js: The runtime environments for executing server-side JavaScript.
5. Styling: Bootstrap for responsiveness and Material UI for professional UI elements.

Development Tools: VS Code (IDE), Postman (API Testing), Git (Version Control).

4. PROJECT DESIGN

4.1 Problem Solution Fit

SB Works design specifically targets the "Lack of Identity" issue. By providing a customizable profile page, freelancers change from being "anonymous numbers" to "professional identities." Sarah can link her specific achievements (HTML, CSS, JS projects) to her profile, which the client can view as soon as she bids.

4.2 Proposed Solution

Our proposed solution is a three-tier architecture:

1. Client Layer (React)
2. Logic Layer (Express/Node)
3. Data Layer (MongoDB)

4.3 Solution Architecture

[FIGURE 4.1: SYSTEM ARCHITECTURE DIAGRAM]

UI Components [Dashboard, Chat, BidList] <---> Axios <---> API Gateway
[Auth, ProjectRoutes, UserRoutes] <---> Mongoose ODM <---> MongoDB
Atlas

This architecture allows for "Separation of Concerns." The design ensures that changes to the database structure don't break the frontend UI components, as the Express layer acts as a strict translation and validation buffer.

5. PROJECT PLANNING & SCHEDULING

The project followed a standard SDLC lifespan over 4 major sprints.

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Phase	Duration	Milestones				
Requirement Gathering	2 Weeks	Journey Map & SRS completion.	Backend Dev	3 Weeks	API endpoints and Mongo Schemas.	Frontend Dev

Progress was tracked via a Trello board, ensuring each Jira ticket for modules like "Bidding Component" was tested before merging.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing was conducted using Postman for unit test validation and Lighthouse for frontend speed.

[FIGURE 6.1: TESTING RESULTS TABLE]

Unit Test: Login Auth -> Result: 200 OK (Pass)

Unit Test: Bid Submit -> Result: Saved to DB (Pass)

Stress Test: 50 concurrent bids -> Latency: 180ms (Pass)

The system was verified to handle up to 100 simultaneous active chat sessions without noticeable degradation in message delivery speeds.

7. RESULTS

7.1 Output Screenshots

[IMAGE 7.1: FREELANCER DASHBOARD VIEW]

Displays Sarah's active projects, bid statistics, and skill badges.

[IMAGE 7.2: PROJECT BIDDING MODAL]

The interactive form where Sarah enters her proposal and attaches samples.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized Platform: No need to switch between apps for project delivery.
- MERN Scalability: The platform grows with the user base.
- Career Building: Dedicated focus on entry-level potential.

Disadvantages

- Real-time dependency: Requires stable internet for chat functionality.
- No Native App: Currently limited to browser-based interactions.

9. CONCLUSION

SB Works has successfully addressed the challenges faced by both clients and freelancers in the modern gig economy. By focusing on a user-centric design (Sarah's story) and a robust technical architecture (MERN stack), the project creates an environment where efficiency and transparency thrive.

The transition from a simple idea to a fully functional platform provides a blueprint for how technical skill and empathetic design can unlock individual professional potential on a global scale.

10. FUTURE SCOPE

The modular nature of SB Works allows for significant future enhancements:

- **AI Integration:** Using NLP (Natural Language Processing) to automatically categorize project descriptions for clients.
- **Payment Gateway:** Integrating Stripe or PayPal to automate project milestones and escrow services.
- **Mobile Extension:** Developing a React Native application to allow Sarah to respond to bakery bids on the go.

11. APPENDIX

Source Code Repository: <https://github.com/project/sb-works-fullstack>

Dataset Structure: JSON-based MongoDB collections (Users, Projects, Bids, Chats).

```
// Sample Bid Submission Controller exports.createBid =  
(req, res) => { const newBid = new Bid({ projectId:  
req.body.projectId, freelancer: req.user.id, amount:  
req.body.amount, proposal: req.body.proposal });  
newBid.save().then(bid => res.json(bid)); };
```

END OF REPORT